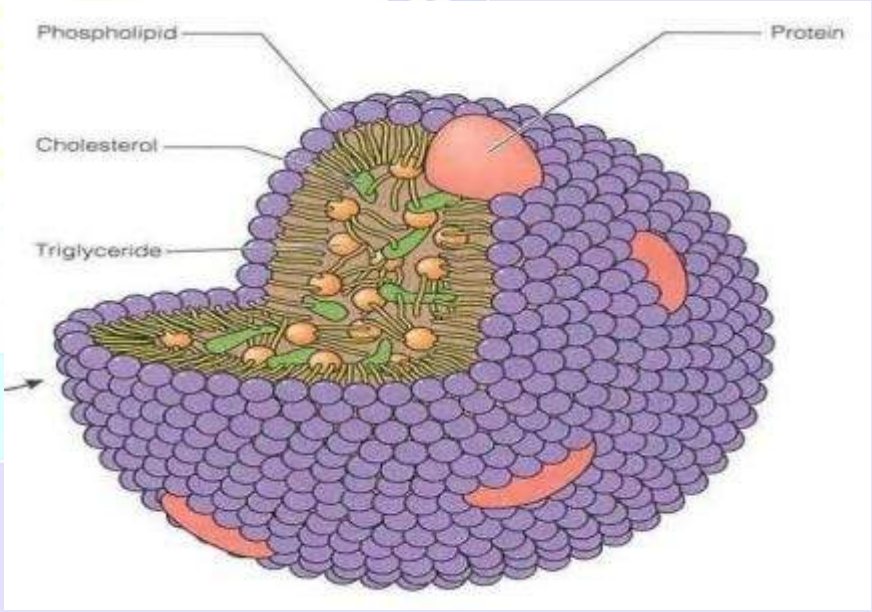
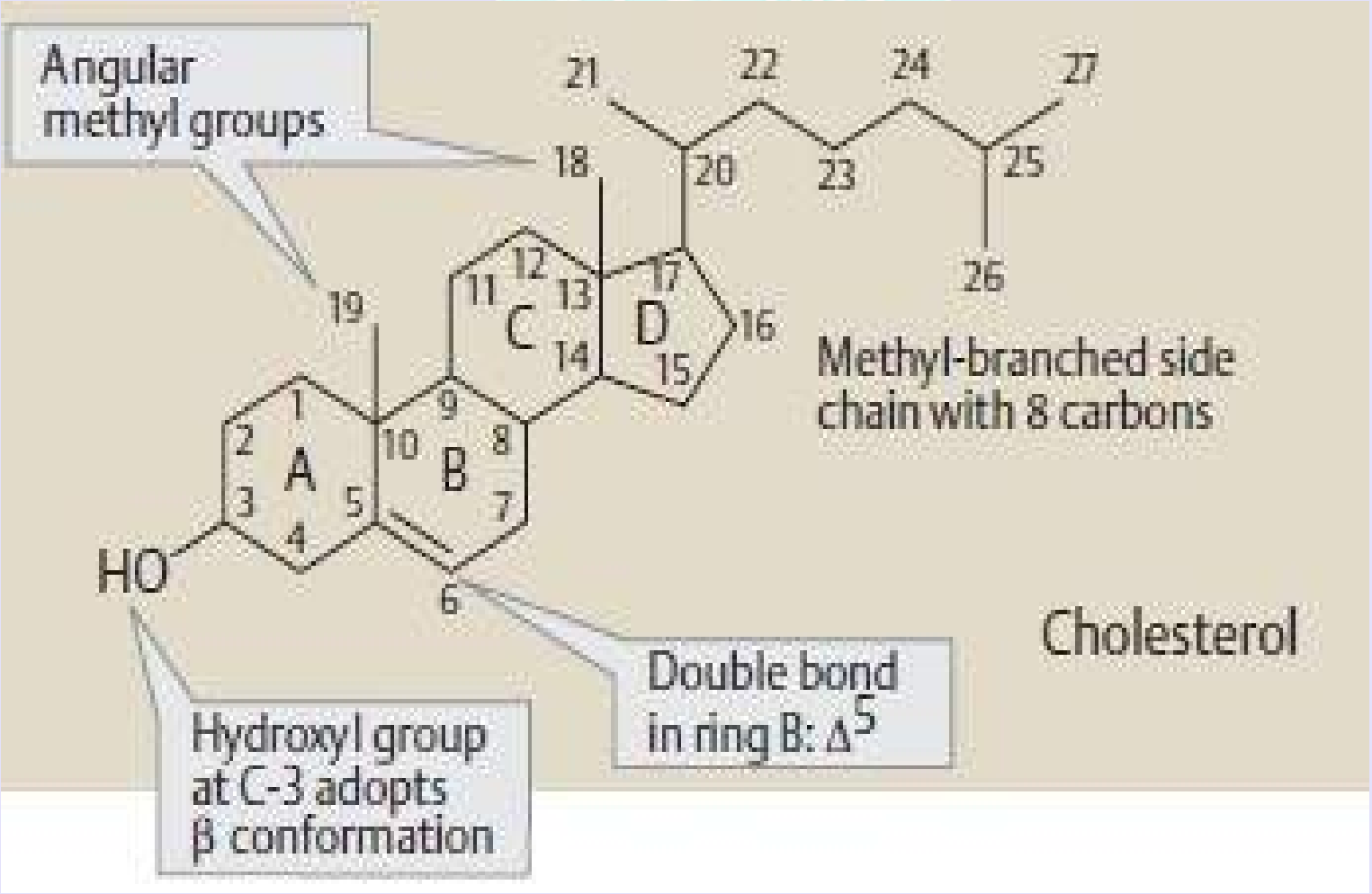
Fate of Dietary Cholesterol and indigenous Cholesterol in human body (Metabolism of Cholesterol)



Objectives

- By the end of the lecture the student should be able to :
- Discuss briefly the structure of cholesterol
- Enlist different enzymes involved in synthesis of cholesterol
- Discuss the different steps involved in cholesterol synthesis
- Discuss the degradation of cholesterol
- Justify the role of HMG CoA reductase as rate limiting enzyme

Structure


Angular methyl groups
Methyl-branched side chain with 8 carbons
Cholesterol
Double bond in ring B: Δ5
Hydroxyl group at C-3 adopts β conformation
HO
A
B
C
D
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
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Cholesterol: Description

- Hydrophobic compound
- Steroid nucleus
- Most plasma cholesterol in an esterified form
- Transported with lipoprotein particle
- Soluble by phospholipids, bile salts
- Coronary artery disease

Synthesis of cholesterol

- All tissues.
- All the C atoms → acetate and NADPH provides the reducing equivalents
- Enzymes for synthesis occur → cytosol, membrane of ER

Steps: Synthesis

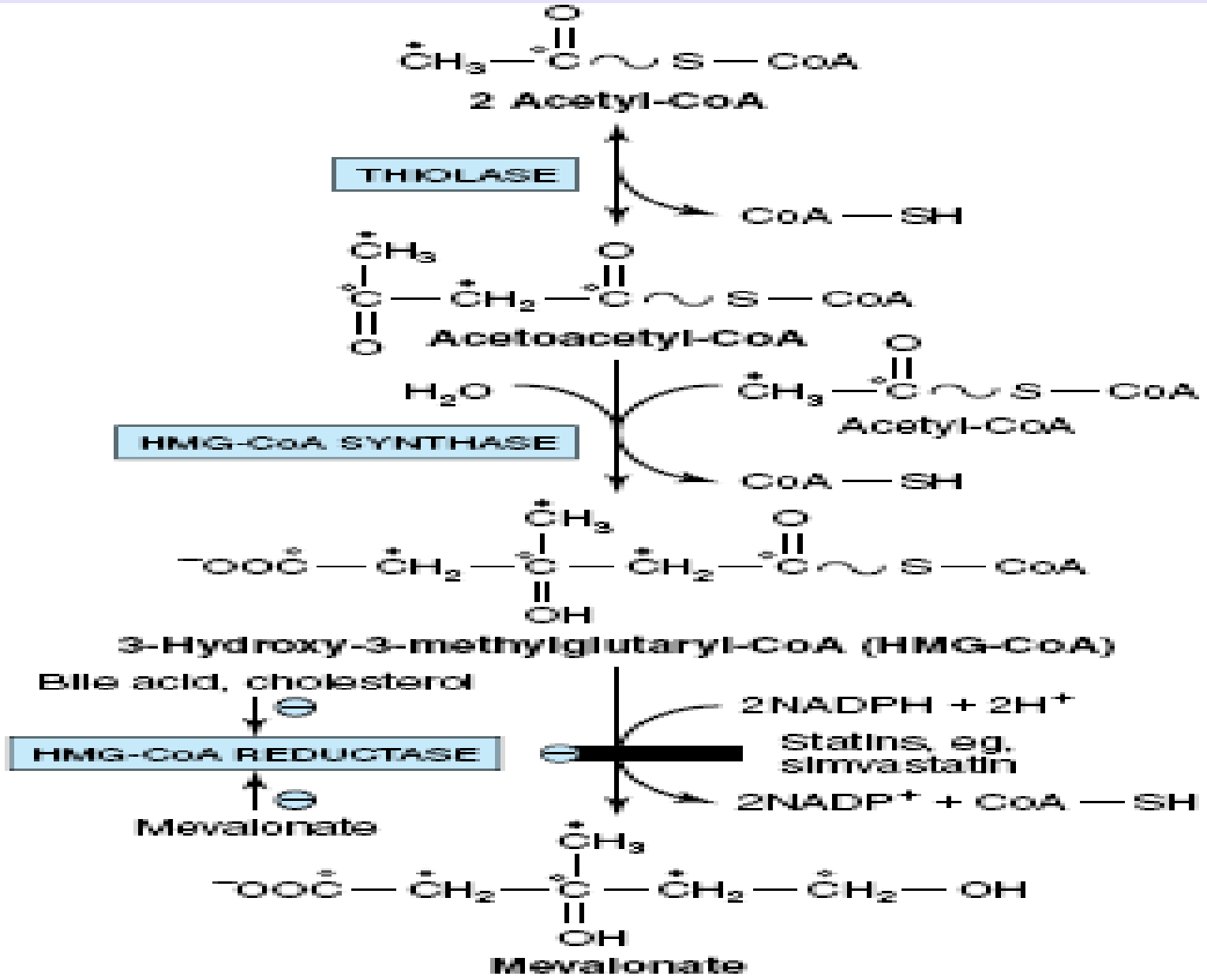
- Synthesis of 3 hydroxy-methylglutaryl CoA (HMG CoA)
- Synthesis of mevalonic acid
- Synthesis of cholesterol

Synthesis of 3 hydroxy-methylglutaryl CoA (HMG CoA)

- First 2 reactions are similar to ketogenesis → HMG CoA
- Liver parenchymal cells contains 2 isozymes → HMG synthase
- Cytosolic enzyme → cholesterol synthesis
- Mitochondrial enzyme → ketone bodies

Synthesis of mevalonic acid

- Cytosol
- HMG CoA reduced to mevalonic acid by HMG CoA reductase
- Irreversible, Rate limiting step, cytosol
- Uses 2 NADPH , release CoA



ure 26–1. Biosynthesis of mevalonate. HMG-CoA

Synthesis of cholesterol

- Mevalonic acid → 5-pyrophosphate →isopentenyl (IPP) →isomerise to 3,3 dimethylallyl pyrophosphate (DPP)
- IPP and DPP condensed → geranyl pyrophosphate (GPP)
- 2nd molecule of IPP condensed with GPP → farnesyl pyrophosphate (FPP)
- 2 molecule FPP → sequalene → sterol lanosterol → cholesterol
- Smith-Lemil-Opitz syndrome → deficiency of 7 dehydro cholesterol 7 reductase (AR disorder)→ multisystem disorder.

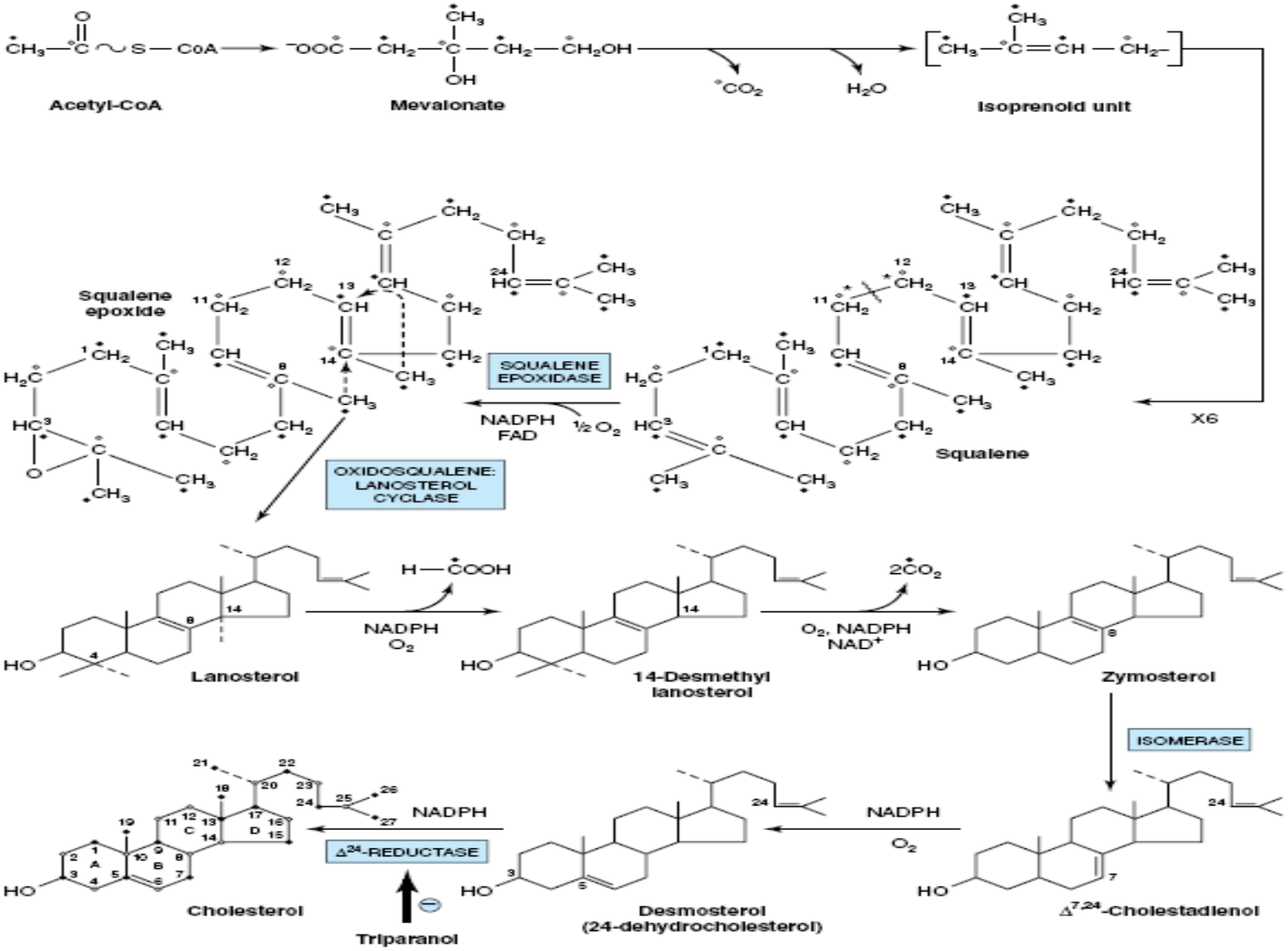
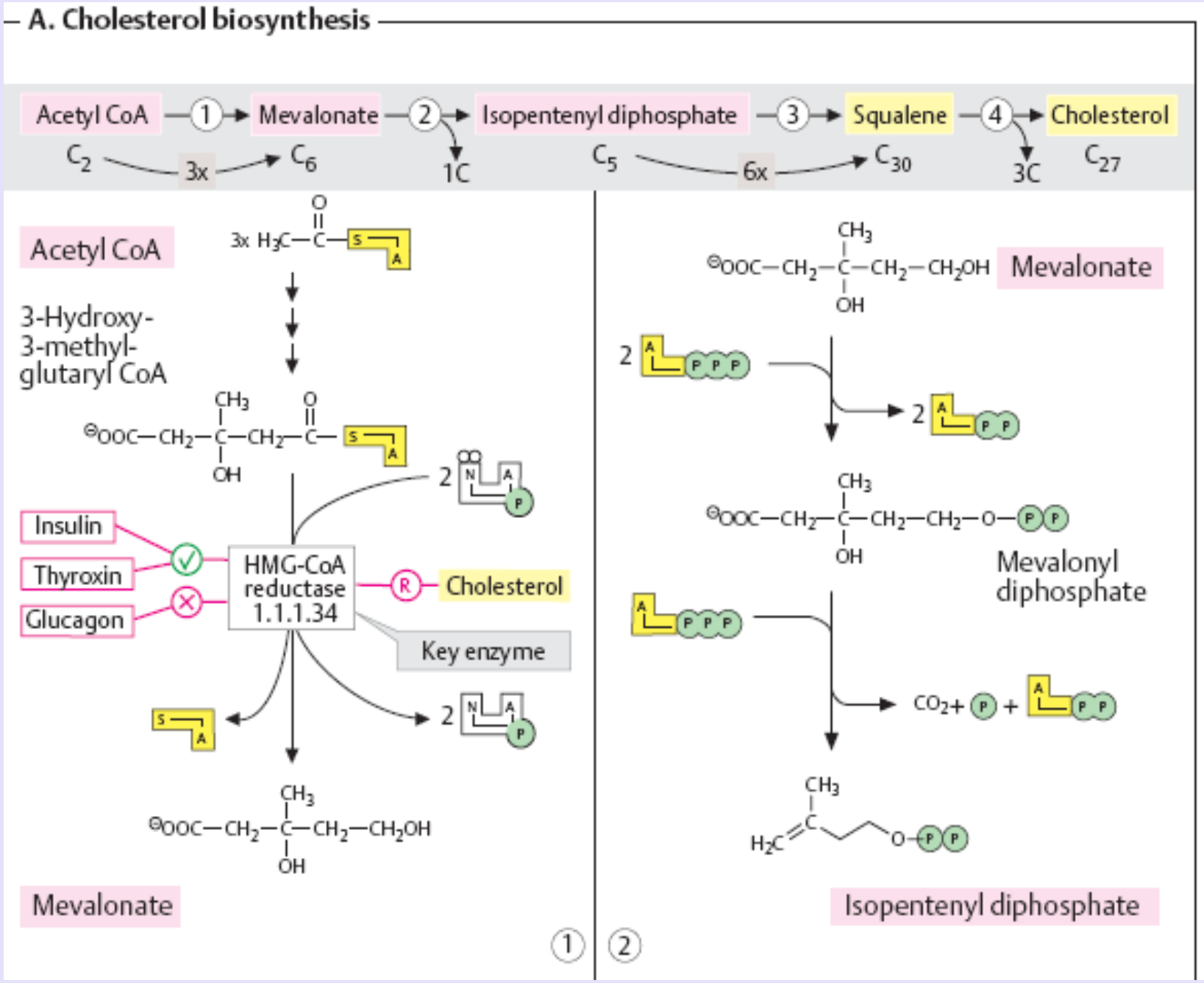
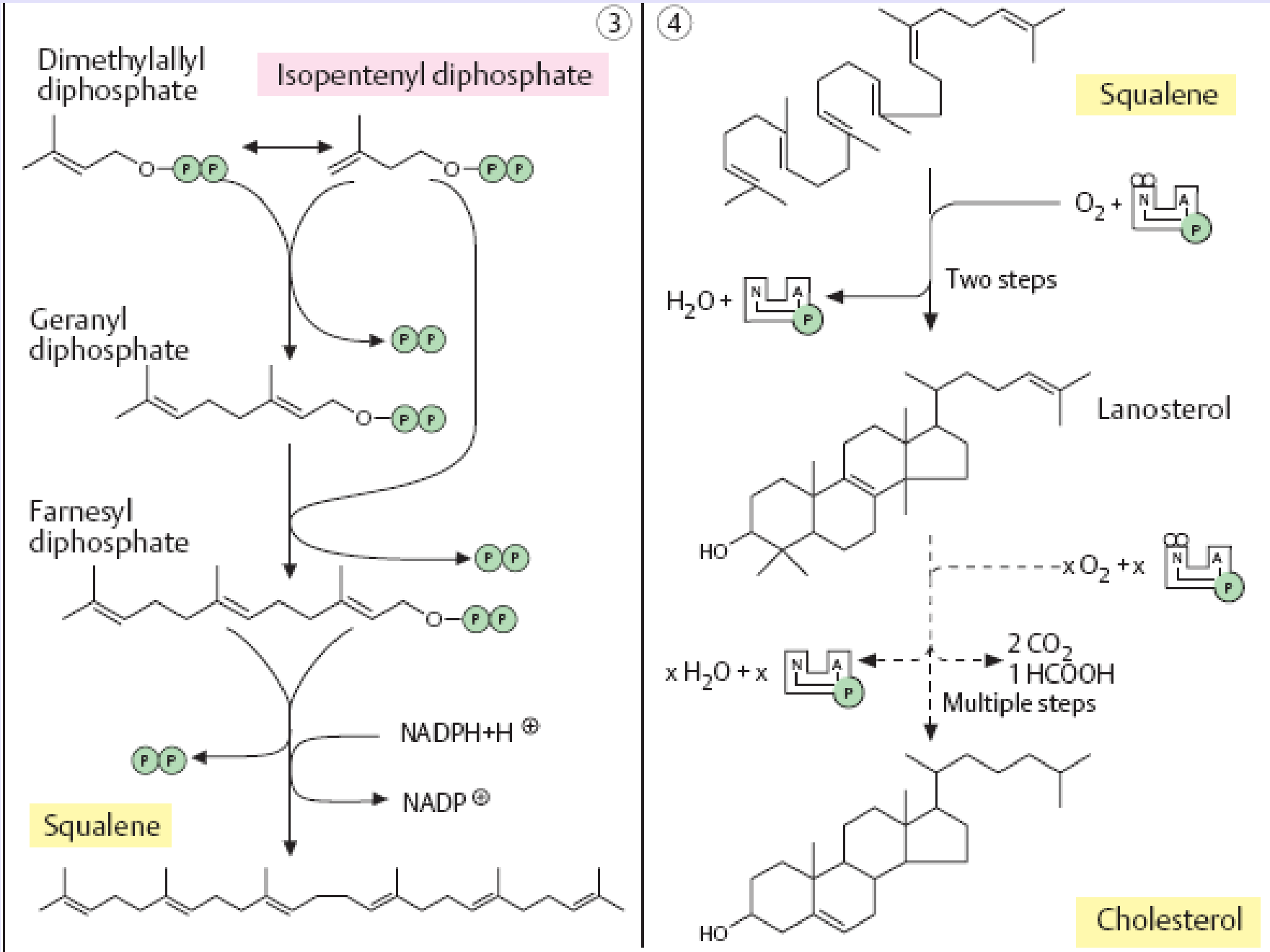


Figure 26–3. Biosynthesis of cholesterol. The numbered positions are those of the steroid nucleus and the open and solid circles indicate the fate of each of the carbons in the acetyl moiety of acetyl-CoA. Asterisks: Refer to labeling of squalene in Figure 26–2.


A. Cholesterol biosynthesis
Acetyl CoA
1
Mevalonate
2
Isopentenyl diphosphate
3
Squalene
4
Cholesterol
C2
3x
C6
1C
C5
6x
C30
3C
C27
Acetyl CoA
3-Hydroxy-
3-methyl-
glutaryl CoA
Insulin
Thyroxin
Glucagon
HMG-CoA
reductase
1.1.1.34
Cholesterol
Key enzyme
Mevalonate
Mevalonate
Mevalonyl
diphosphate
Isopentenyl diphosphate
1
2


3
Dimethylallyl diphosphate
Isopentenyl diphosphate
Geranyl diphosphate
Farnesyl diphosphate
NADPH+H ⊕
NADP ⊕
Squalene
4
Squalene
O_2 +
H_2O +
Two steps
Lanosterol
HO
x O_2 +x
x H_2O + x
2 CO_2
1 HCOOH
Multiple steps
HO
Cholesterol

Degradation of Cholesterol

- In humans → cholesterol ring is not metabolized → CO_2 & H_2O

Degradation of Cholesterol

- Sterol nucleus → bile acid and bile salts → excreted in feces
- Intestinal bacteria
- Isomers Coprostanol and cholestanol
- Bulk of neutral fecal sterols

Reference

- Lippincott’s biochemistry 5th edi page 219